

AgentHack Challenge

Autonomous AI Sales Agent

Build an AI sales system that understands a company, finds the right customers, researches them, sells the right service, manages responses, and follows up automatically.

Core idea: RAG is only the starting point. The final system should behave like an autonomous sales team.

Input	Company PDF or complete company description + ICP information
Core flow	RAG -> ICP -> Lead Discovery -> Filtering -> Deep Research -> Qualification -> Outreach -> Follow-up
Required memory	Short-term memory + long-term memory
Required communication	Email; meeting scheduling; follow-up
Competition focus	Correct decisions, useful research, personalization, reliability, and working automation

1. Challenge Overview

Build an **AI-powered sales system** that first understands a company from a provided PDF or text description, then uses that knowledge to discover, research, qualify, contact, and manage potential customers.

The system should not be a simple chatbot. It should operate as a complete sales pipeline and maintain context throughout the customer lifecycle.

2. Company Knowledge

The user must be able to provide the company information in either of two ways:

- Upload the provided company PDF.
- Paste or enter the complete company description as text.

That information should be processed into a RAG / knowledge layer so the AI can reliably understand:

- What the company sells
- Services and capabilities
- Target industries and customers
- Case studies and previous work
- Pricing or packages when available

- Technologies and integrations
- Business limitations or conditions

The system should use this knowledge when deciding what to sell to each prospect.

3. Define the Ideal Customer Profile (ICP)

The user should define the target customer before lead generation. At minimum, ask for:

- Target location
- Target industry / market
- Company size or other basic qualification criteria
- What the user wants to target specially (problem, use case, service, or type of customer)

The system may convert these answers into a structured ICP that is used throughout the pipeline.

4. Lead Discovery

The agent should use a staged process rather than deeply researching every company immediately:

Search -> Cheap Filtering -> Potential Leads -> Deep Research -> Qualification

Cheap filtering should eliminate obviously irrelevant companies before expensive or detailed research is performed.

5. Deep Company Research

For the most promising companies, research as much useful business information as the available tools permit:

- Company profile and website
- Employees and important departments
- News and recent events
- Funding and growth signals
- Technology / tools / integrations
- Possible business problems or buying signals
- Other evidence useful for qualification

6. Lead Qualification and Confidence

For each researched company, the system should decide whether it is a strong lead and show a clear confidence score.

Example: ABC Logistics - 92% Potential Lead

The AI should explain why it gave that score. Useful factors can include ICP fit, problem fit, service fit, buying signals, company size, location, and evidence quality.

The system must also be able to conclude that a company is **not qualified**. The objective is to identify the best leads, not to maximize the number of contacts.

7. Service Matching

The AI must determine which product or service from the original company is most appropriate for each qualified prospect.

Example: if a prospect has a large volume of WhatsApp customer inquiries, the system should recognize when a WhatsApp AI chatbot or customer-support automation is a stronger recommendation than another service.

The recommendation should be grounded in the company knowledge stored in the RAG.

8. Decision-Maker Identification

For qualified companies, the system should identify relevant decision-makers and important stakeholders, such as:

- CEO / Founder / Co-Founder
- CTO
- CFO
- COO
- VP Sales / Head of Sales
- Head of Support
- Other role relevant to the recommended service

The AI should prioritize the people who are most relevant to the actual service being sold.

9. Personalized Outreach

Generate personalized emails for the selected contacts. The message should consider the person's role, the company's situation, the likely problem, and the best matching service.

Personalization must be evidence-based. The system should avoid inventing facts about a prospect.

Each contact should receive a message suited to their role. For example, a CTO-focused message may emphasize integrations and technical fit, while a sales leader may care more about productivity and revenue impact.

10. Email and Meeting Link

The system should send the outreach email through the provided email mechanism or simulation. When appropriate, the email should contain a meeting / booking link.

Each outreach action should be stored with the company, contact, role, lead score, recommended service, message, evidence, and timestamp.

11. Response Classification

When a prospect replies, an agent should classify the response, for example:

- Positive / Interested
- Meeting requested
- Question
- Pricing objection
- Technical objection
- Not interested
- Not now
- Wrong person / Referral

- Other

The classification should determine the next action in the sales pipeline.

12. Follow-Up

If there is no response, the system should automatically maintain the lead and schedule a follow-up.

Day 0 -> Email #1 -> Wait 3 days -> Follow-up Agent -> Email #2

When a company replies positively and a meeting is finalized, the agent should reply with the new meeting link and the fixed meeting time.

13. Meeting and Admin Automation

Once a meeting is finalized with a prospect, the system should manage the meeting workflow.

- Generate or provide the meeting link with the finalized fixed time.
- Send the meeting details to the admin through WhatsApp when the meeting is finalized.
- Store the meeting information in memory.
- Remind the admin 30 minutes before the meeting.
- Include a short meeting briefing in the reminder: customer problem, recommended service, important objections/questions, and key points to discuss.

14. Memory

The system must maintain both short-term and long-term memory.

Short-term memory: current research, current conversation, current task, current meeting context, and active agent state.

Long-term memory: company history, contacts, emails, replies, objections, lead scores, follow-ups, meetings, and final sales status.

15. Sales Pipeline

A simple pipeline view should show where each prospect currently stands:

Discovered -> Potential -> Researching -> Qualified -> Contacted -> Interested -> Meeting Scheduled -> Converted

The system should also support negative outcomes such as Not Qualified, Not Interested, and Do Not Contact.

16. End-to-End Workflow

Company PDF / Text -> RAG -> ICP -> Lead Discovery -> Cheap Filtering -> Deep Research -> Qualification -> Lead Score -> Decision Makers -> Service Matching -> Personalized Email -> Response Classification -> Follow-Up -> Meeting Scheduling -> Memory -> Sales Pipeline

The goal is to create an end-to-end autonomous sales system, not merely a document Q&A; chatbot.

17. Required Features Summary

Company knowledge	PDF or text input; RAG-based company understanding
ICP	Location, industry, company criteria, special targeting
Lead discovery	Search, cheap filtering, potential leads

Deep research	Company, people, news, funding, technology, signals
Qualification	Lead confidence percentage + explanation
Decision makers	Major relevant people + contact details when available
Sales	Role-specific personalized email
Responses	Classification and next action
Follow-up	3-day follow-up + second email
Meetings	Fixed time + meeting link
Memory	Short-term + long-term
Pipeline	Persist and update lead state throughout the lifecycle

18. Extra Features

Teams may implement the following extra features for additional credit:

- New meeting link generated when sending the email to the company.
- WhatsApp integration.
- If the company has a WhatsApp number, the agent also messages the company on WhatsApp.
- The agent reminds the admin about the meeting and also sends the meeting link on WhatsApp in the reminder.

19. Evaluation

The recommended scoring is:

Area	Weight
Recommended features working perfectly	70%
Extra features	15%
Frontend	15%

The overall submission score is then divided between the presentation and the code:

Video Presentation - 60% Code - 40%

Judges should focus on whether the required workflow actually works, whether the agent makes sensible decisions, whether information is grounded in the provided company data, and whether the system maintains context through the complete sales lifecycle.

20. Submission Requirements

All three submissions are mandatory:

1	Live Demo Video	Maximum 1.5 minutes. Demonstrate the working project and the key end-to-end workflow.
2	Code Explanation Video	Maximum 1 minute. Explain the architecture, agents, RAG, memory, tools, and important technical decisions.

3	GitHub Repository	Submit the GitHub repository link. Make sure to include an .env.example file in the repository.
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Final objective: Build an autonomous AI sales system that can understand the company, discover the right leads, research them, decide what to sell, contact the right people, understand responses, schedule meetings, remember the relationship, and follow up.